

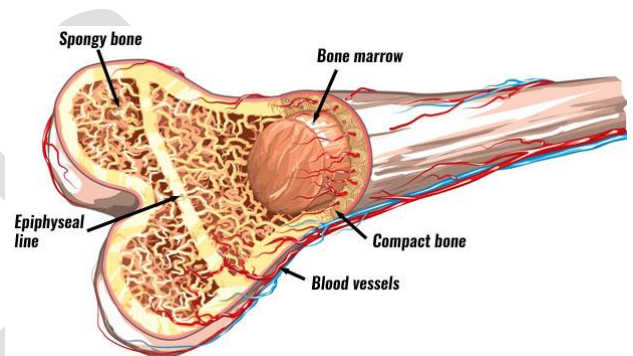
Bone (Osseous Tissue)

Bone is a strong and durable type of connective tissue. It is composed chiefly of calcium phosphate and calcium carbonate. It also serves as a storage area for calcium, playing a large role in calcium balance in the blood. The bone also consists of cells and inters cellular substance or matrix. The matrix of bones has the special property of being rigid because it is impregnated with mineral salts, mainly calcium phosphate. Hence, the chief properties of the bone tissue is its toughness and hardness.

Bone is made up of cells, fibers (Collagen) and ground substances.

Simply define as:

Bone is a special type of connective tissue impregnated with minerals having physical properties of being tough and hard.



Bone Cells:

Three different cells type are found in osseous tissue:

- 1- Osteoblast (bone formation)
- 2- Osteocytes (mature bone cell)
- 3- Osteoclast (Bone Resorption)

Bone Resorption:

It is a process in which osteoclasts break down the tissue in bone and release minerals to blood.

Bone matrix:

The bone matrix consists of inorganic and organic components. It is responsible for hardness of bone tissue and constitutes about 65% of dry weight of bone. Calcium and phosphate are chief bone minerals, but substantial quantities of sodium, magnesium, carbonate and citrate are also found. Calcium and phosphate occur as excess.

Types of bone:

Two types of bone tissue can be seen in cross section of a bone.

1- Compact bone (Hard): compact bone are dense having no intercellular spaces and cavities.

2- Spongy bone (Soft): is a area which on microscopic examination shows intercellular spaces and cavities.

Microscopic structure of the bone tissue:

The most characteristic feature of bone tissue is its lamellar structure. The bone matrix is arranged as layers or lamellae ranging from 3 to 7 μm in thickness. The collagenous fibers of lamella run parallel to each other.

Classification of bones:

- 1- Long bone (Humerus)
- 2- Short bone (Carpals)
- 3- Irregular bone (skull and vertebrae)
- 4- Flat bone (Ribs, parietal, sternum)
- 5- Sesamoid bone (Patella)

Head => Epiphyses

Shaft => Diaphysis

Functions:

- ⇒ Posture and shape
- ⇒ Minerals reservoir
- ⇒ Place for hematopoieses
- ⇒ Body movement

Joints

A structure in human body at which two parts of skeleton are fitted together.

Or

A joint is the site in which two or more bones articulates or come together.

Types of joints:

1. Fibrous or fixed joints
2. Cartilaginous or slightly movable joints
3. Synovial or freely movable joints

Fibrous or fixed joints:

These are immovable joints closely packed with dense connective tissue having no cavities. Example: Skull bone, teeth, maxilla, mandible.

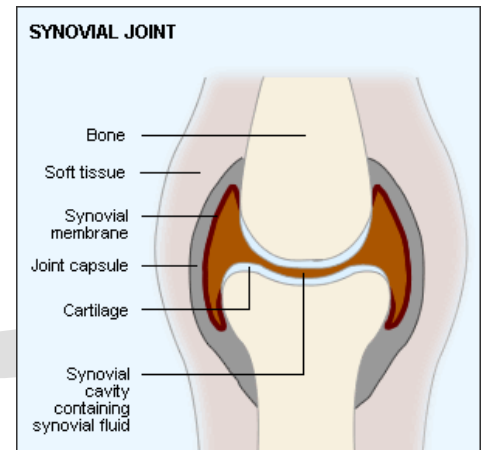
Cartilaginous or slightly movable joints:

There is a pad of fibrocartilage between the ends of the bones that form joints which allow the very slight movement where the pads of cartilage is compressed. Example: Spine, Ribs.

Synovial or freely movable joints:

A **synovial joint** is the type of **joint** found between bones that move against each other, such as the **joints** of the limbs (e.g. shoulder, hip, elbow and knee).

Synovial membrane (or **synovium**) – a special layer of cells that lines the **joint** capsule and produces the **synovial** fluid.

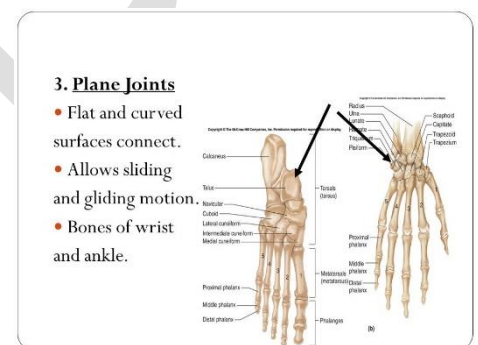


Types of synovial joints

There are six types of synovial joints as:

1- Plane, gliding or sliding joints:

A gliding or plane joint formed between bones that meet flat or nearly flat surfaces known as plane joints. Example: Carpal and Tarsals.



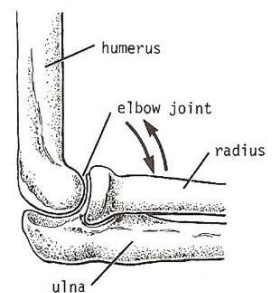
2- Hinge joints:

Hinge joints are formed between two or more bone, where bones can move along on axis. For example: Elbow, knee, ankle.

Movements: Flexion and extension.

3- Pivot joints:

Pivot joints are those which allow movement around single axis (one bone can rotate about another bone). For example: Proximal Radioulnar joint and Distal Radioulnar joint.



4-Condylloid joints:

Condylloid joint is ovoid articular surface that is receive into an elliptical cavity. For example: Wrist joint, Thumb, Radiocarpal joint.



Movements: Flexion and Extension, Abduction and adduction, circumduction.

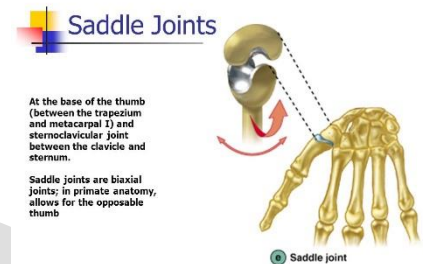
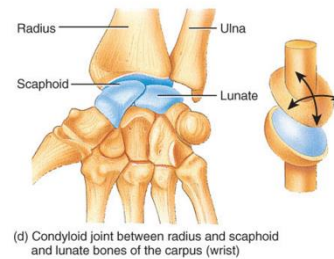
5-Saddle joint:

Two bones are concave on one side and convex on the other side that fit together allowing biaxial movement. For example: Carpometacarpal joint.

Movements: Flexion and Extension, Abduction and adduction, circumduction.

6-ball and socket joint:

The head of one bone articulates in the socket of another bone. For example: Shoulder and hip joint.



BALL AND SOCKET JOINTS

Movement in all three planes

Shoulder and hip joints are ball and socket. This type of joint is **multiaxial**.

